An 80 kg astronaut: same mass, different weight



Earth

$g = 9.8\ m/s^2$

m = **80 kg**

W = 784 N



Moon

$g = 1.6\ m/s^2$

m = **80 kg**

W = 128 N



Mars

$g = 3.7\ m/s^2$

m = **80 kg**

W = 296 N



Jupiter

$g = 24.8\ m/s^2$

m = **80 kg**

W = 1984 N

Mass (m): amount of matter, in **kg**. It is the same everywhere.

Weight ($W = m \cdot g$): a force, in **N**. It depends on the planet.